



Joint Velocity-Growth Flow Matching for Single-Cell Dynamics Modeling

Dongyi Wang¹, Yuanwei Jiang¹, Zhenyi Zhang², Xiang Gu¹, Peijie Zhou², Jian Sun¹

¹ School of Mathematics and Statistics, Xi'an Jiaotong University, China

² School of Mathematical Sciences, Peking University, Beijing, China

NEURAL INFORMATION
PROCESSING SYSTEMS

Background & Motivation

Single-cell RNA seq yields
unpaired and **unbalanced** data.

Simulation-based models (e.g., TIGON [1], DeepRUOT [2])

■ Growth rate 🙄

■ Scalability 🙄

Simulation-free models (e.g., OT-CFM [3], SF2M [4])

■ Growth rate 🙄

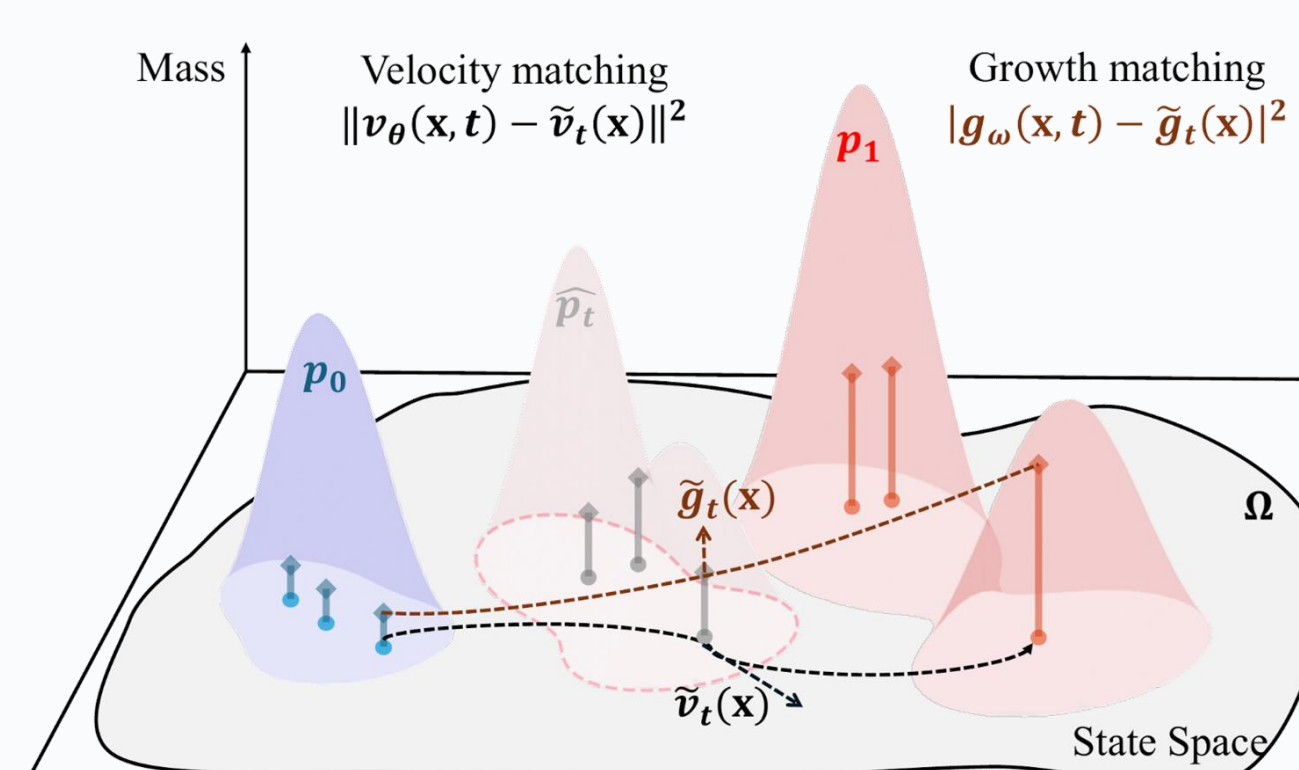
■ Scalability 🙄

Motivation:

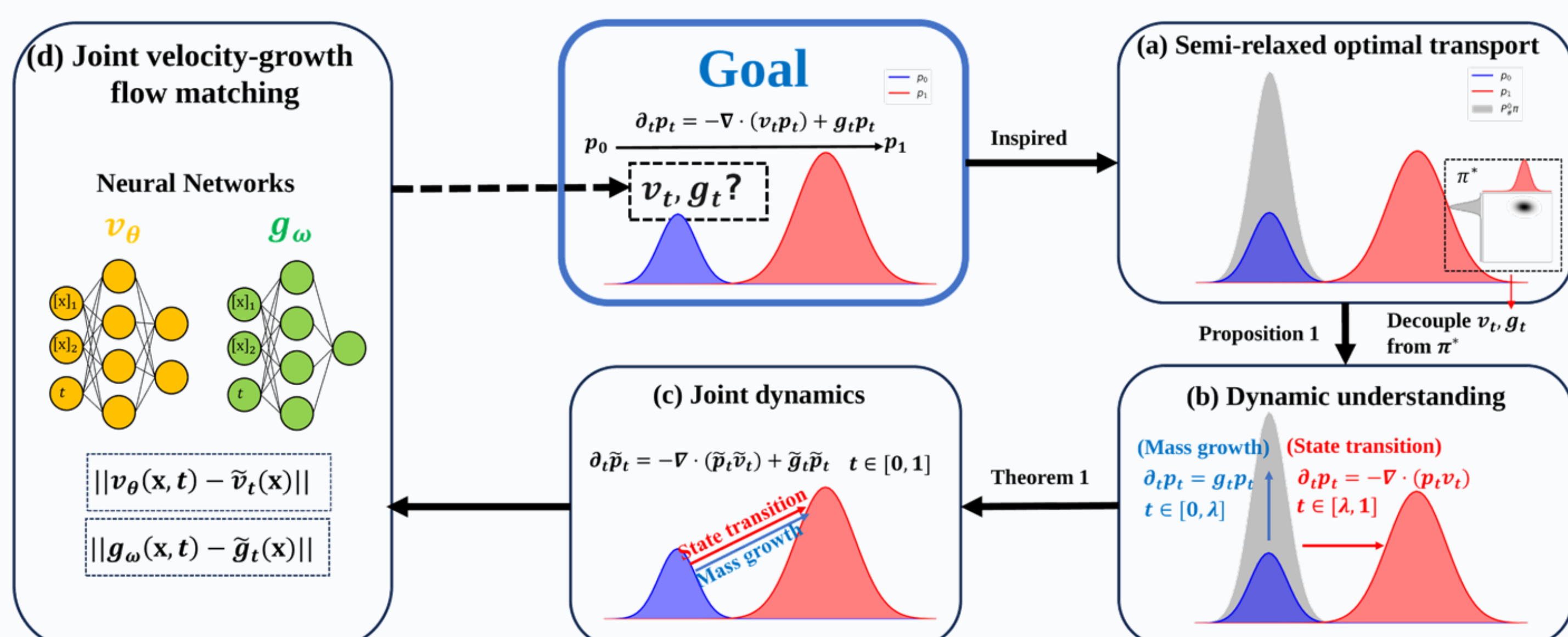
Develop a model which jointly learns the **state transition** and **mass growth** of single-cell evolution by **flow matching**.

■ Growth rate 🙄

■ Scalability 🙄



Velocity-Growth Flow Matching via Semi-relaxed OT



1. Goal: find v_t, g_t between unbalanced p_0, p_1 .

2. **Two-period** dynamic understanding of semi-relaxed OT (SOT) (a,b).

3. Build a novel **joint dynamic** process ($\tilde{v}_t, \tilde{g}_t$) between unbalanced distributions based on dynamic SOT (c).

4. Regress $\tilde{v}_t, \tilde{g}_t$ using neural networks (d).

➤ Two-period dynamic understanding of SOT (a,b)

Static formulation

$$\min_{\pi \geq 0} \mathcal{J}_{\text{SOT}}(\pi) \triangleq \int_{\Omega^2} c(\mathbf{x}_0, \mathbf{x}_1) d\pi(\mathbf{x}_0, \mathbf{x}_1) + \text{KL}(\mathcal{P}_{\#}^0 \pi \| p_0) \quad \text{s.t.} \quad \mathcal{P}_{\#}^1 \pi = p_1$$

- The two formulations share the same optimal objective function.
- The optimal objective function value of the two-period problem is independent of λ .
- v_t^*, g_t^* can be constructed using π^* .

Dynamic formulation

$$\min_{(v_t, g_t) \in \mathcal{C}_{\lambda}(p_0, p_1)} \mathcal{J}_{\text{SOT}}^{\lambda}(v_t, g_t) \triangleq (1 - \lambda) \int_{\Omega} \int_{\Omega} p_t(\mathbf{x}) \|v_t(\mathbf{x})\|^2 d\mathbf{x} + \mathcal{H}(g_t, p_t)$$

$$\mathcal{H}(g_t, p_t) = \int_{\Omega} p_0(\mathbf{x}) (e^{\int_0^{\lambda} g_t(\mathbf{x}) dt} (\int_0^{\lambda} g_t(\mathbf{x}) dt - 1) + 1) d\mathbf{x}$$

$$\mathcal{C}_{\lambda}(p_0, p_1) = \{(v_t, g_t) : \partial_t p_t = g_t p_t, t \in [0, \lambda]; \partial_t p_t = -\nabla \cdot (p_t v_t), t \in (\lambda, 1]\}$$

➤ Building joint unbalanced dynamics (c)

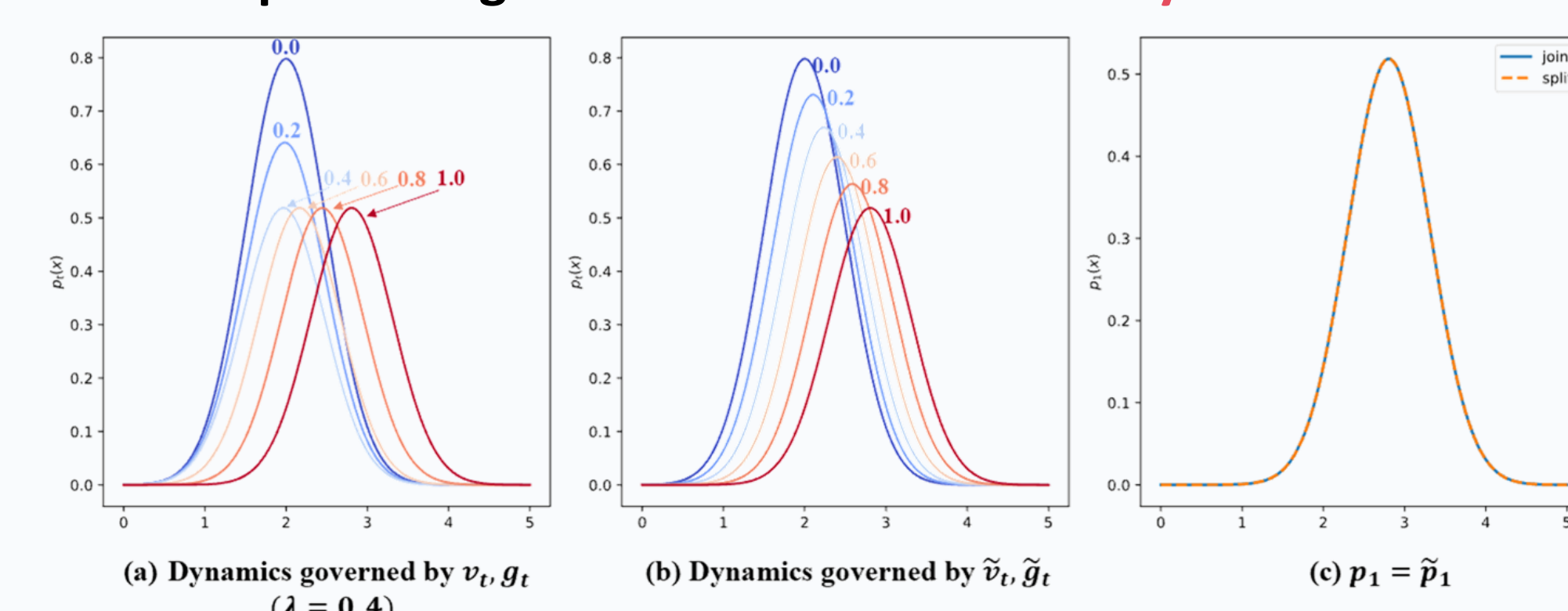
The two-period evolution does **not align** with the behavior observed in biological systems. We want transport and growth occur **simultaneously**.

Given the same initial p_0 , the two-period dynamics and joint dynamics ends in the same target distribution p_1 .

Two-period dynamics $\partial_t p_t = g_t p_t, t \in [0, \lambda]; \quad \partial_t p_t = -\nabla \cdot (p_t v_t), t \in (\lambda, 1]$

Joint dynamics $\partial_t \tilde{p}_t = -\nabla \cdot (\tilde{p}_t \tilde{v}_t) + \tilde{g}_t \tilde{p}_t, t \in [0, 1]$

$$\tilde{v}_t(\mathbf{x}) = (1 - \lambda) \cdot v_{(1-\lambda)t+\lambda}(\mathbf{x}) \quad \tilde{g}_t(\mathbf{x}) = \lambda \cdot g_{\lambda t}(\psi_{\tilde{v}_t}^{-1}(\mathbf{x}))$$



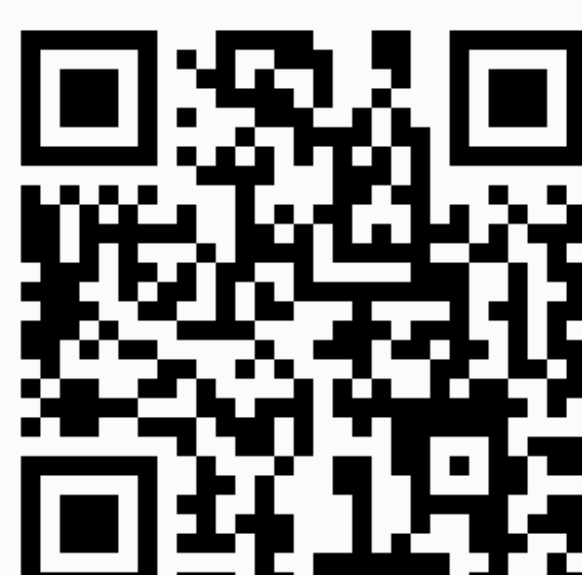
➤ Regress $\tilde{v}_t, \tilde{g}_t$ using neural networks (d)

$$\mathcal{L}_{\text{VGFM}}(\theta, \omega) = \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_1) \sim \pi^*} \mathbb{E}_t \left[\|v_{\theta}(\mathbf{x}_t, t) - (\mathbf{x}_1^j - \mathbf{x}_0^i)\|^2 + |g_{\omega}(\mathbf{x}_t, t) - \log([\pi^* \mathbf{1}_m]_i)|^2 \right]$$

Incorporate few-steps distribution fitting loss (Wasserstein distance) to improve performance:

$$\mathcal{L}(\theta, \omega) = \mathcal{L}_{\text{VGFM}}(\theta, \omega) + \mathcal{L}_{\text{OT}}(\theta, \omega)$$

Code

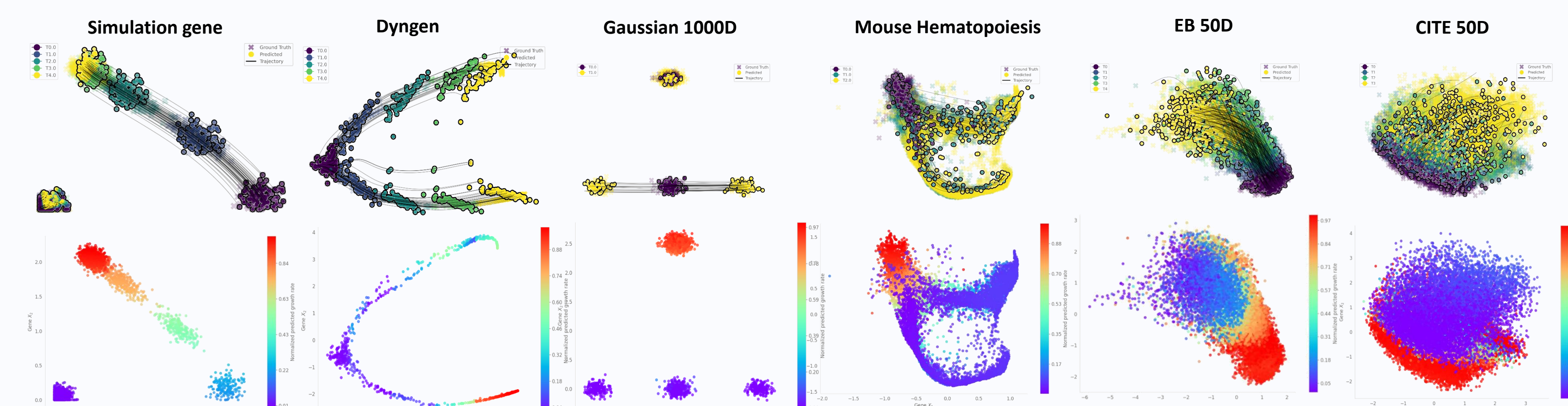


Paper

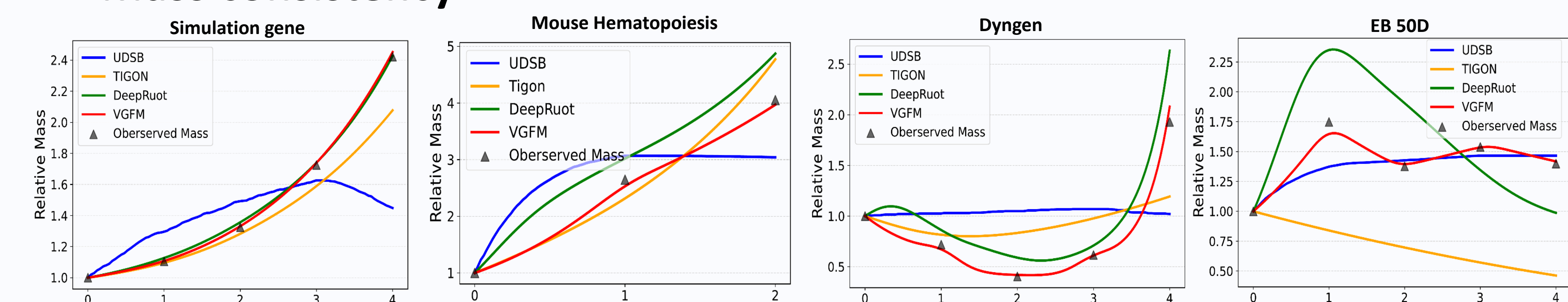


Experiments

➤ Trajectory and growth reconstruction



➤ Mass consistency



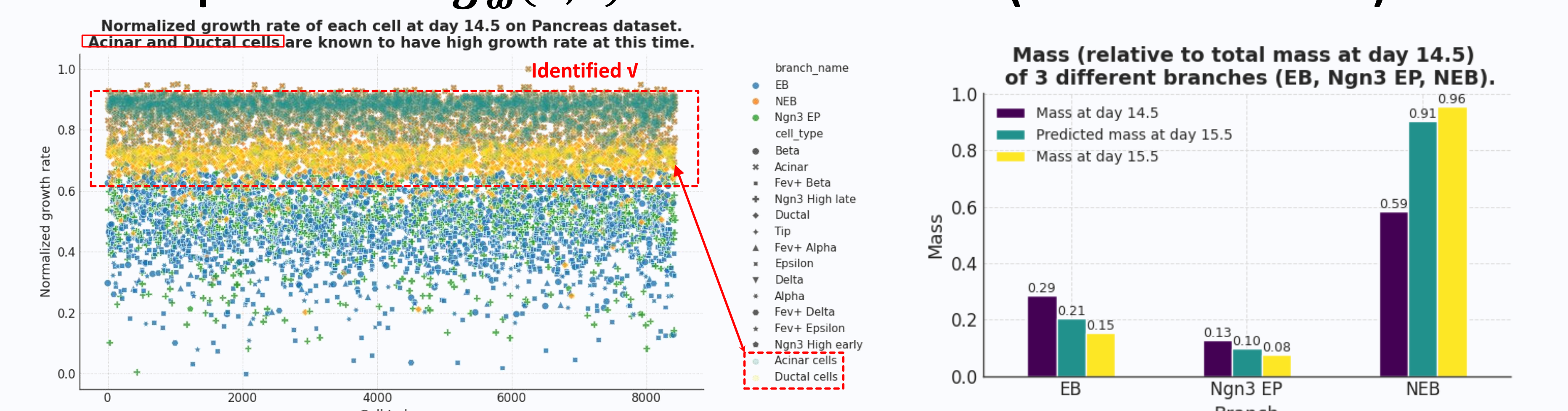
➤ Wasserstein-1 distance & Relative Mass Error

Method	Simulation Gene (2D)		Dyngen (5D)		Gaussian (1000D)	
	$\mathcal{W}_1$	RME	$\mathcal{W}_1$	RME	$\mathcal{W}_1$	RME
OT-CFM* [22]	0.302	—	3.926	—	10.126	—
OT-MFM* [38]	0.311	—	3.976	—	11.008	—
UDSB [57]	0.665	0.192	1.914	0.658	N/C	N/C
TIGON [23]	0.099	0.065	1.029	0.542	N/C	N/C
DeepRUOT [26]	0.068	0.016	0.474	0.199	N/C	N/C
VGFM	0.046	0.006	0.420	0.053	3.010	0.037

➤ Hold out experiment

Method	EB		CITE
	5D	5D	50D
OT-CFM [22]	0.790	0.882	38.756
SF ² M [41]	0.793	0.920	38.524
UDSB [57]	1.206	2.023	44.168
OT-MFM [38]	0.713	0.724	36.394
DeepRUOT [26]	0.774	0.845	38.681
VGFM	0.676	0.745	37.386

➤ Interpretation of $g_{\omega}(x, t)$: Pancreas dataset (2000-dimensions)



VGFM Successfully reconstructed cell type-wise growth rate **without being given any prior knowledge** of the cell types.

[1] Sha Y., Qiu Y., Zhou P., Nie Q. *Reconstructing growth and dynamic trajectories from single-cell transcriptomics data.* Nature Machine Intelligence, 6(1):25–39, 2024.

[2] Zhang Z., Li T., Zhou P. *Learning stochastic dynamics from snapshots through regularized unbalanced optimal transport.* ICLR, 2025.

[3] Tong A., Fatras K., Malkin N., Huguet G., Zhang Y., Rector-Brooks J., Wolf G., Bengio Y. *Improving and generalizing flow-based generative models with minibatch optimal transport.* TMLR, 2024.

[4] Tong A.Y., Malkin N., Fatras K., Atanackovic L., Zhang Y., Huguet G., Wolf G., Bengio Y. *Simulation-free Schrödinger bridges via score and flow matching.* AISTATS, 2024.